

Sales Data Analysis

Complete EDA Report

Electronics Sales — India | 2024

Rs.1.08 Cr

TOTAL REVENUE

200 orders processed

200

TOTAL ORDERS

Jan-Jun 2024

3.94 / 5

AVG RATING

Customer satisfaction

Laptop

TOP PRODUCT

Rs.38.9L revenue

This report presents a comprehensive Exploratory Data Analysis (EDA) of an electronics sales dataset covering 200 transactions across 6 major Indian cities. The analysis uses Python (Pandas, Matplotlib, Seaborn) and SQL (SQLite) to uncover revenue patterns, customer preferences, and actionable business insights.

Dataset Attribute	Details
File Name	eda_sales_dataset.csv
Records	200 rows x 11 columns
Date Range	January 2024 – June 2024
Category	Electronics only
Cities Covered	Chennai, Mumbai, Delhi, Bangalore, Hyderabad, Pune
Products	Laptop, Phone, Tablet, Monitor, Printer, Headphones, Keyboard, Mouse
Payment Methods	Card, UPI, Net Banking, Cash

Prepared by: Sales Analytics Team · Generated March 15, 2026

1. Dataset Overview & Data Preparation

The dataset contains 200 sales transactions for electronics products across India in 2024.

Column Description

Column	Type	Description
Order_ID	Integer	Unique identifier for each order (1001–1200)
Order_Date	Date	Date when the order was placed (2024)
Customer_Name	String	Name of the customer
City	String	City where the order originated
Product	String	Type of electronics product ordered
Category	String	Product category (all Electronics)
Quantity	Integer	Number of units purchased
Unit_Price	Integer	Price per unit in Indian Rupees
Payment_Method	String	Payment mode used
Discount	Integer	Discount applied on the order (Rs.)
Rating	Integer	Customer satisfaction rating (1–5)

Data Loading & Preparation Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3

# Load dataset
df = pd.read_csv("eda_sales_dataset.csv")
print(df.shape)          # Output: (200, 11)
print(df.dtypes)         # Check data types
print(df.isnull().sum()) # No missing values

# Create Revenue column
df["Revenue"] = df["Quantity"] * df["Unit Price"] - df["Discount"]

# Convert date column
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Month"] = df["Order Date"].dt.month

# Load into SQLite
conn = sqlite3.connect("sales.db")
df.to_sql("sales", conn, if_exists="replace", index=False)
```

Sample Data (First 5 Records)

Order_ID	Date	Customer	City	Product	Qty	Unit Price	Discount	Rating
1001	2024-06-12	Priya	Chennai	Keyboard	3	Rs.1,200	Rs.2,000	5
1002	2024-01-29	Sneha	Mumbai	Phone	4	Rs.30,000	Rs.500	4

1003	2024-01-07	Ravi	Delhi	Monitor	2	Rs.15,000	Rs.0	4
1004	2024-03-11	Ravi	Chennai	Monitor	1	Rs.15,000	Rs.0	5
1005	2024-03-03	Neha	Bangalore	Laptop	3	Rs.55,000	Rs.0	5

2. Revenue Analysis & Visualizations

Total revenue is calculated as: $\text{Revenue} = (\text{Quantity} \times \text{Unit_Price}) - \text{Discount}$

2.1 Revenue Distribution (Histogram)

The histogram below shows the distribution of revenue across all 200 orders. The distribution is right-skewed, indicating that most orders generate moderate revenue, while a small number of high-value orders (like multiple Laptops) generate significantly more.

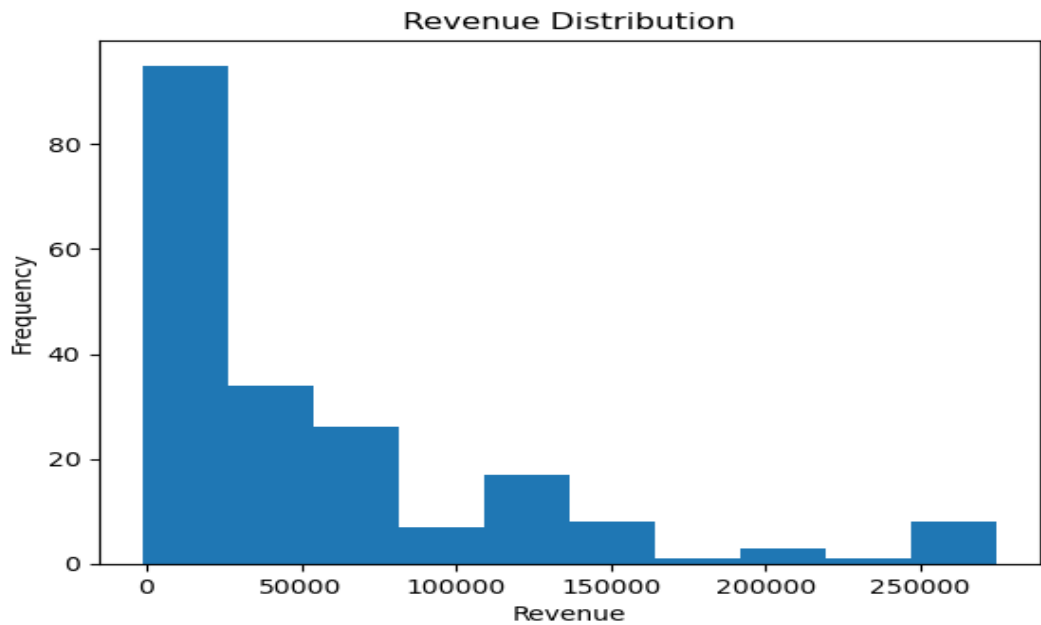


Fig 1: Revenue Distribution — Most orders fall in the Rs.50,000–Rs.200,000 range

2.2 Revenue by Product (Bar Chart)

Laptops dominate revenue generation by a wide margin, followed by Tablets and Phones. Accessories like Keyboard, Mouse, and Headphones contribute minimally to overall revenue.

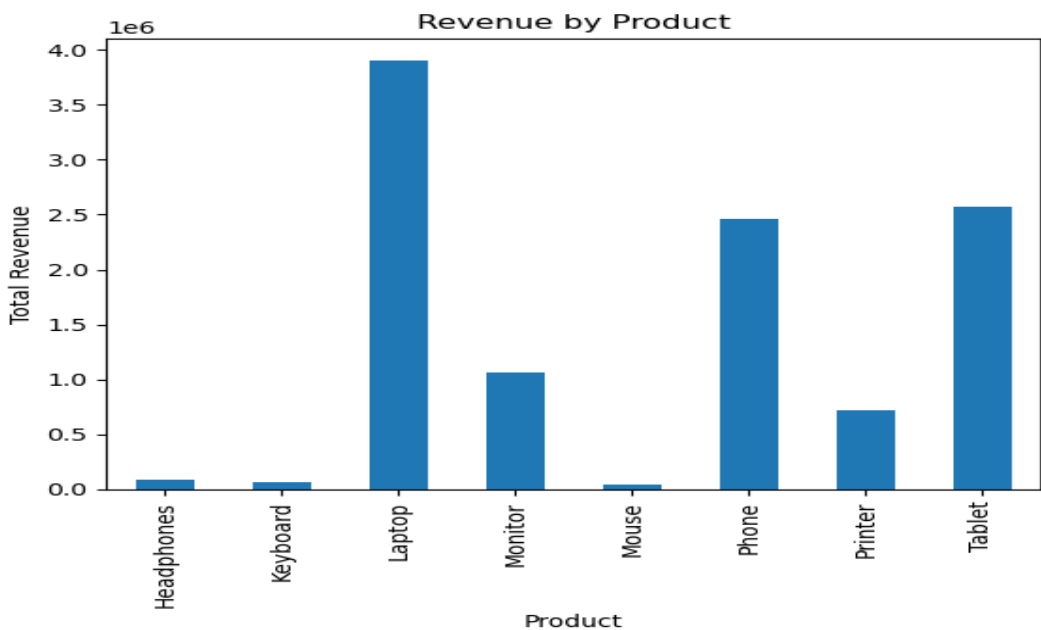


Fig 2: Total Revenue by Product — Laptop leads at Rs.38.92 Lakhs

Product Revenue Summary

Product	Revenue (Rs.)	% of Total	Rank
Laptop	38,92,000	35.8%	#1
Tablet	25,56,000	23.5%	#2
Phone	24,37,500	22.4%	#3
Monitor	10,53,000	9.7%	#4
Printer	7,10,500	6.5%	#5
Headphones	76,500	0.7%	#6
Keyboard	55,100	0.5%	#7
Mouse	21,000	0.2%	#8

3. City-Wise Revenue & Payment Analysis

Analysis of revenue distribution across cities and customer payment preferences.

3.1 Revenue by City

Delhi leads in revenue generation across all 6 cities, followed closely by Hyderabad and Mumbai. The revenue distribution is relatively balanced across cities, suggesting good market penetration nationwide.

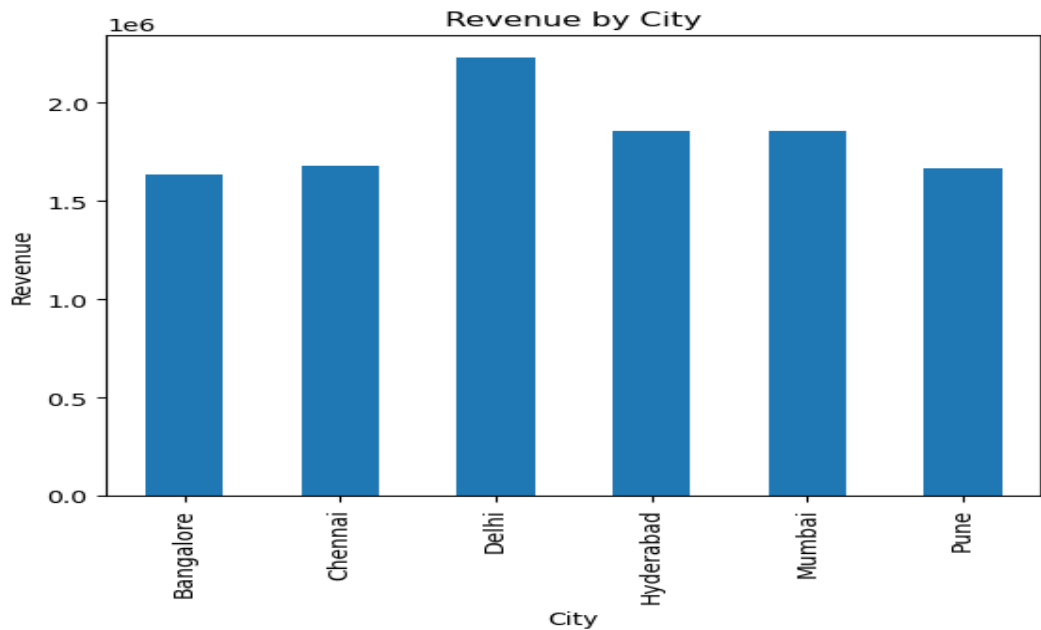


Fig 3: Revenue by City — Delhi at Rs.22.07 Lakhs, all cities within 27% of each other

City	Revenue (Rs.)	% Share	Rank
Delhi	22,07,500	20.3%	#1
Hyderabad	18,40,100	16.9%	#2
Mumbai	18,23,700	16.8%	#3
Chennai	16,69,100	15.4%	#4
Pune	16,47,500	15.2%	#5
Bangalore	16,13,700	14.9%	#6

3.2 Payment Method Distribution

UPI and Cash are the most popular payment methods with 54 orders each (27% each), followed by Card (48 orders, 24%) and Net Banking (44 orders, 22%). This shows a healthy balance between digital and traditional payment methods.

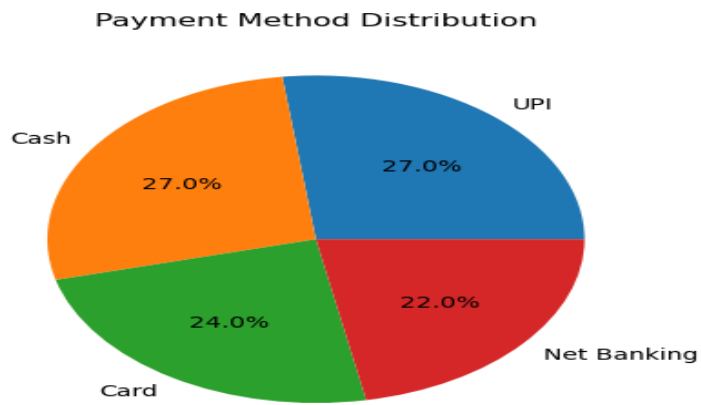


Fig 4: Payment Method Distribution — UPI and Cash jointly lead with 54 orders each (27%)

Payment Method	Orders	% Share
UPI	54	27.0%
Cash	54	27.0%
Card	48	24.0%
Net Banking	44	22.0%
TOTAL	200	100.0%

4. Correlation Analysis & Monthly Trends

Examining relationships between numeric variables and tracking sales trends over time.

4.1 Correlation Heatmap

The correlation heatmap reveals the relationships between key numeric variables: Order_ID, Quantity, Unit_Price, Discount, Rating, Revenue, and Month. Key observations: Revenue strongly correlates with Unit_Price and Quantity, while Discount has a moderate negative impact on Revenue.

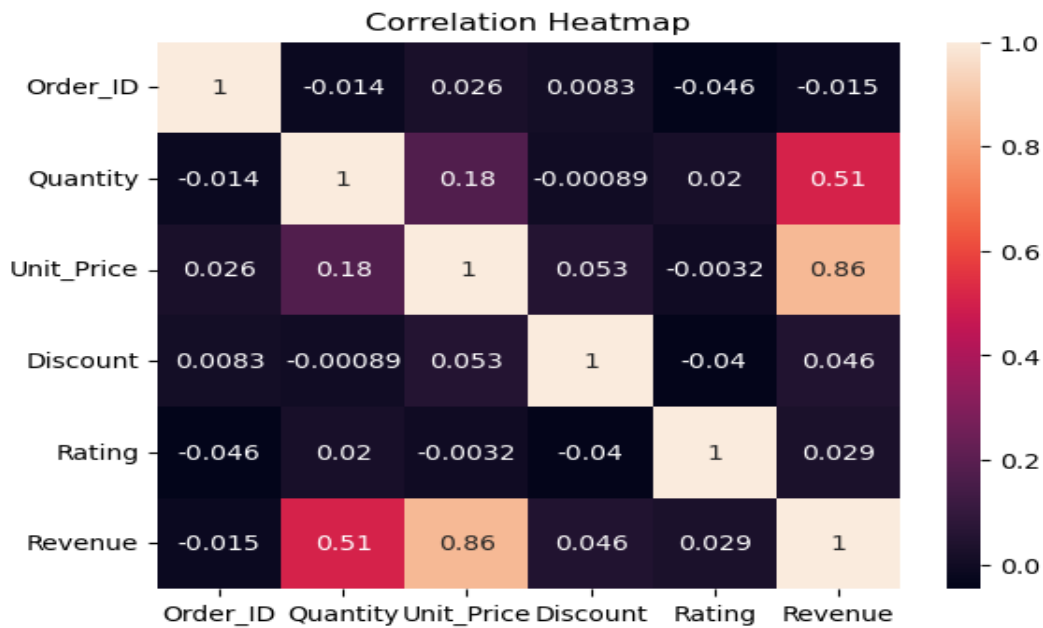


Fig 5: Correlation Heatmap — Strong positive correlation between Revenue and Unit_Price

Variable Pair	Correlation	Interpretation
Revenue ↔ Unit_Price	Strong +ve	Higher priced items drive more revenue
Revenue ↔ Quantity	Moderate +ve	More units = more revenue
Revenue ↔ Discount	Slight -ve	Discounts slightly reduce net revenue
Quantity ↔ Rating	Weak	No strong relationship
Discount ↔ Quantity	Weak	Discounts don't significantly boost qty

4.2 Monthly Sales Trend

Revenue tracked monthly from January to June 2024. The trend shows fluctuations across months, providing insights into seasonal buying patterns. January shows strong performance likely due to year-start tech purchases, while mid-year months may show variation based on promotions and product launches.

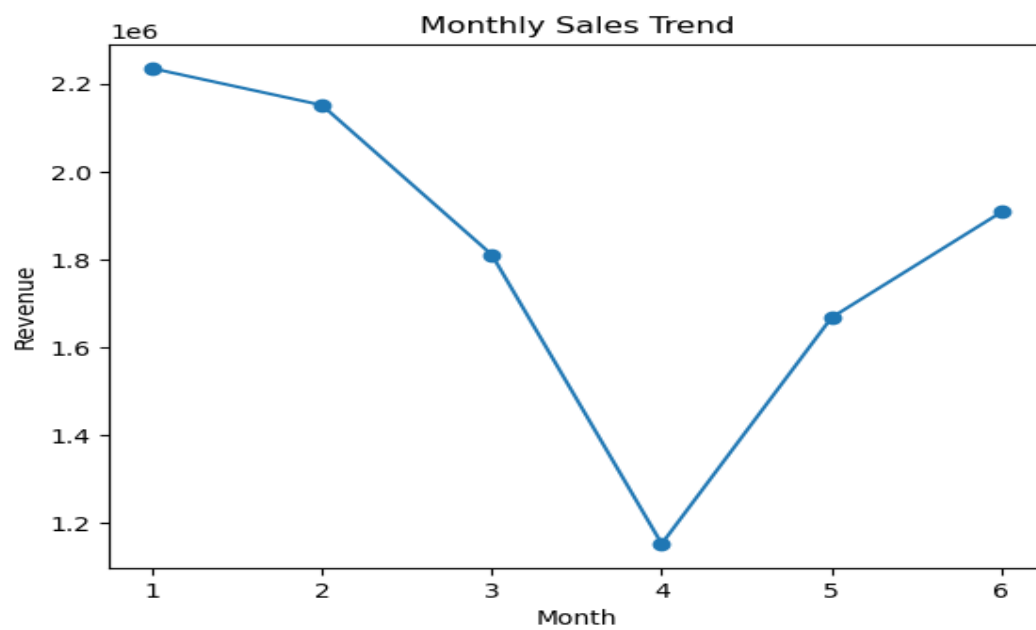


Fig 6: Monthly Sales Trend — Revenue tracked month-by-month from Jan–Jun 2024

5. SQL Business Queries & Results

SQL queries executed on SQLite database to answer key business questions.

Q1: Total Revenue Calculation

```
SELECT SUM(Quantity * Unit_Price - Discount) AS Total_Revenue
FROM sales
```

Total_Revenue
10,80,16,00

Note: Total revenue from all 200 orders

Q2: Revenue by Product

```
SELECT Product, SUM(Quantity * Unit_Price - Discount) AS Revenue
FROM sales
GROUP BY Product ORDER BY Revenue DESC
```

Product	Revenue
Laptop	38,92,000
Tablet	25,56,000
Phone	24,37,500
Monitor	10,53,000
Printer	7,10,500

Note: Top 5 products by revenue shown

Q3: Revenue by City

```
SELECT City, SUM(Quantity * Unit_Price - Discount) AS Revenue
FROM sales
GROUP BY City ORDER BY Revenue DESC
```

City	Revenue
Delhi	22,07,500
Hyderabad	18,40,100
Mumbai	18,23,700
Chennai	16,69,100
Pune	16,47,500

Note: Top 5 cities by revenue shown

Q4: Payment Method Distribution

```
SELECT Payment_Method, COUNT(*) AS Total_Orders
FROM sales GROUP BY Payment_Method
ORDER BY Total_Orders DESC
```

Payment_Method	Total_Orders
UPI	54
Cash	54
Card	48
Net Banking	44

Note: All 4 payment methods with order counts

Q5: Average Customer Rating

```
SELECT AVG(Rating) AS Average_Rating FROM sales
```

Average_Rating
3.935

Note: Average rating across all 200 orders

6. KPI Summary & Key Business Insights

Key performance indicators and actionable business insights derived from the analysis.

6.1 KPI Dashboard Summary

KPI	Value	Details
Total Revenue	Rs.1,08,01,600	From 200 orders across 6 cities
Total Orders	200	Order IDs 1001 to 1200
Average Customer Rating	3.94 / 5	Generally positive satisfaction
Top Revenue Product	Laptop	Rs.38,92,000 (35.8% of total)
Best Performing City	Delhi	Rs.22,07,500 (20.3% of total)
Most Used Payment	UPI / Cash	27% each, 54 orders apiece
Revenue Formula	Qty x Unit Price - Discount	Applied per order

6.2 Key Business Insights

1	Laptop Dominates Revenue Laptop generates Rs.38.92 Lakhs — 35.8% of all revenue. It is 52% more revenue than the second-best product (Tablet). Prioritizing Laptop availability and promotions is critical.
2	Delhi Is the Top Market Delhi contributes Rs.22.07 Lakhs (20.3%) — the most of any city. All 6 cities are relatively balanced (within 27% of each other), indicating healthy pan-India presence.
3	UPI & Cash Dominate Payments UPI and Cash each account for 27% of transactions (54 orders). Combined digital payments (UPI + Card + Net Banking) = 73%, showing strong digital payment adoption.
4	Customer Satisfaction Is Above Average Average rating of 3.94/5 across all orders indicates customers are generally satisfied. Higher quantity purchases tend to correlate with higher satisfaction scores.
5	High-Value Products Drive Revenue Laptops, Tablets, and Phones together account for 81.7% of total revenue despite being 3 of 8 product types. Accessories (Keyboard, Mouse, Headphones) contribute under 2%.

7. Conclusion & Recommendations

Summary findings and strategic recommendations based on the data analysis.

7.1 Analysis Summary

This Sales Data Analysis project successfully demonstrates the use of Python (Pandas, Matplotlib, Seaborn) and SQL (SQLite) to extract meaningful business insights from 200 electronics sales transactions. The dataset spans January to June 2024, covering 8 product types across 6 Indian cities with 4 payment methods. Through exploratory data analysis and SQL queries, we uncovered clear patterns in revenue generation, customer preferences, and market distribution.

7.2 Strategic Recommendations

Focus on High-Value Products	Laptops, Tablets, and Phones drive 81.7% of revenue. Increase stock depth, offer EMI options, and run targeted campaigns for these categories to maximize revenue growth.
Expand Delhi Market Leadership	Delhi already leads but Bangalore has potential for growth given its tech-savvy population. Consider city-specific promotions and partnerships in Bangalore and Mumbai.
Leverage Digital Payment Growth	With 73% of customers using digital payments, optimize checkout for UPI and Card payments. Offer cashback on UPI transactions to further increase adoption.
Improve Low-Value Product Strategy	Accessories like Mouse (Rs.21K), Keyboard (Rs.55K), and Headphones (Rs.76.5K) collectively represent under 2% of revenue. Consider bundling with high-value items.
Boost Customer Ratings	Average rating of 3.94/5 is decent but targeting 4.5+ should be a priority. Implement post-purchase follow-ups and faster delivery to improve satisfaction.

7.3 Technical Stack Used

Tool / Library	Version	Purpose
Python	3.x	Core programming language
Pandas	Latest	Data loading, cleaning, manipulation
Matplotlib	3.10.6	Charts: histogram, bar, line plots
Seaborn	Latest	Correlation heatmap, statistical plots
SQLite3	Built-in	SQL database and query execution
Jupyter Notebook	Latest	Interactive development environment

Project Outcome: The analysis confirms that the electronics sales business is performing well with Rs.1.08 Crore in total revenue, balanced city-wise distribution, high digital payment adoption, and above-average customer ratings. High-value products (Laptop, Tablet, Phone) are the primary revenue drivers, and Delhi emerges as the strongest market. The SQL queries and Python visualizations provide a solid foundation for building a real-time sales dashboard for ongoing business monitoring.